Microbiology and Infectious Disease

A lytic transglycosylase connects bacterial focal adhesion complexes to the peptidoglycan cell wall

Reviewed Preprint
v1 • July 16, 2024
Not revised

Carlos Ramirez Carbo, Olalekan G Faromiki, Beiyan Nan

Department of Biology, Texas A&M University, College Station, TX 77843, USA • The Genetics and Genomics Interdisciplinary Program, Texas A&M University, College Station, TX 77843, USA

Abstract

The Gram-negative bacterium *Myxococcus xanthus* glides on solid surfaces. Dynamic bacterial focal adhesion complexes (bFACs) convert proton motive force from the inner membrane into mechanical propulsion on the cell surface. It is unclear how the mechanical force transmits across the rigid peptidoglycan (PG) cell wall. Here we show that AgmT, a highly abundant lytic PG transglycosylase homologous to *Escherichia coli* MltG, couples bFACs to PG. Coprecipitation assay and single-particle microscopy reveal that the gliding motors fail to connect to PG and thus are unable to assemble into bFACs in the absence of an active AgmT. Heterologous expression of *E. coli* MltG restores the connection between PG and bFACs and thus rescues gliding motility in the *M. xanthus* cells that lack AgmT. Our results indicate that bFACs anchor to AgmT-modified PG to transmit mechanical force across the PG cell wall.

eLife assessment

The manuscript by Carbo et al. reports a novel role for the MltG homolog AgmT in gliding motility in M. xanthus. The authors provide **convincing** data to demonstrate that AgmT is a cell wall lytic enzyme (likely a lytic transglycosylase), its lytic activity is required for gliding motility, and that its activity is required for proper binding of a component of the motility apparatus to the cell wall. The findings are **valuable** as they contribute to our understanding of the molecular mechanisms underlying the interaction between gliding motility and the bacterial cell wall.

https://doi.org/10.7554/eLife.99273.1.sa2

Introduction

In natural ecosystems, the majority of bacteria attach to surfaces[1]. Surface-associated motility is critical for many bacteria to navigate and populate their environments[2]. The Gram-negative bacterium *Myxococcus xanthus* moves on solid surfaces using two independent mechanisms: social (S-) motility and adventurous (A-) motility. S-motility, analogous to the twitching motility in *Pseudomonas* and *Neisseria*, is powered by the extension and retraction of type IV pili[3,4]. A-motility, also known as gliding motility, does not depend on conventional motility-related cell surface appendages, such as flagella or pili[2,5]. In *M. xanthus*, AglR, AglQ, and AglS form a membrane channel that functions as the gliding motor by harvesting proton motive force[6,7]. The motor unit associates with at least fourteen gliding-related proteins that reside in the cytoplasm, inner membrane, periplasm, and outer membrane[8–10].

Recently, the mechanisms for bFAC assembly and the coupling between bFACs and substrate begin to emerge[8,11,12]. In each gliding cell, two distinct populations of gliding complexes coexist: dynamic complexes move along helical paths and static complexes remain fixed relative to the substrate[11,13–15]. Motors transport incomplete gliding complexes along helical tracks but form complete, force-generating complexes at the ventral side of cells that appear fixed to the substrate (**Fig. 1**)[11,15]. Based on their functional analogy with eukaryotic focal adhesion sites, these complete gliding machineries are called bacterial focal adhesion complexes (bFACs). Despite their static appearance, bFACs are dynamic under single-molecule microscopy. Individual motors frequently display move-stall-move patterns, in which they move rapidly along helical trajectories, join a bFAC where they remain stationary briefly, then move rapidly again toward the next bFAC or reverse their moving directions[13,16,17]. bFACs adhere to the gliding substrate through an outer membrane adhesin[12]. As motors transport bFACs toward lagging cell poles, cells move forward but bFACs remain static relative to the gliding substrate.

bFAC assembly can be quantified at nanometer resolution using the single-particle dynamics of gliding motors[13,18]. Single particles of a fully functional, photoactivatable mCherry (PAmCherry)-labeled AglR display three dynamic patterns, stationary, directed motion, and diffusion[13,18]. The stationary population consists of the motors in fully assembled bFACs, which do not move before photobleach. In contrast, the motors moving in a directed manner carry incomplete gliding complexes along helical tracks, whereas the diffusing motors assemble to even less completeness[11,13,15,18]. As motors only generate force in static bFACs[11], the population of nonmotile motors indicates the overall status of fully assembled bFACs.

However, how bFACs transmit force across the peptidoglycan (PG) cell wall is unclear. PG is a mesh-like single molecule of crosslinked glycan strands that surrounds the entire cytoplasmic membrane. The rigidity of PG defines cell shape and protects cells from osmotic lysis[19,20]. If bFACs physically penetrate PG, their transportation toward lagging cell poles would tear PG and trigger cell lysis. To avoid breaching PG, an updated model proposes that there are two subcomplexes in each bFAC, one on each side of the PG layer. Rather than forming stable and rigid interactions, these two subcomplexes only associate transiently to form a bFAC[11,21] (**Fig. 1**).

As motors reside in the fluid inner membrane, to transmit force to the cell surface, they must push against two relatively rigid structures, one on each side of the membrane, in opposite directions[21] (**Fig. 1**). MreB is a bacterial actin homolog that associates with the motors and plays essential roles in gliding[2,5,6,14,18,22]. The motors could push against MreB filaments in the cytoplasm, whereas PG could be the structure that motors push against in the periplasm[11,21]. The interaction between the inner complex and PG not only satisfies the physical requirement for force generation but also supports efficient bFACs assembly. Without this interaction, once the inner subcomplex disassociates from the outer one, it will quickly escape bFACs and no longer contribute to force generation[21]. In this case, the fraction of time each motor generates force (remains in bFACs) will be extremely low[21]. How the inner complex interacts with PG remains unknown. It was speculated that gliding motors in the inner complex could bind PG directly[11]. However, such binding has not been confirmed by experiments.

In this study, we found that AgmT, a lytic transglycosylase (LTG) for PG, is essential for *M. xanthus* gliding motility. AgmT is required for maintaining PG integrity under the stress from the antibiotic mecillinam. Using single-particle tracking microscopy and coprecipitation assays, we found that AgmT is required for the gliding motors to connect to PG and stall in bFACs. Importantly, expressing *Escherichia coli* MltG heterologously rescues the connection between PG and bFACs and

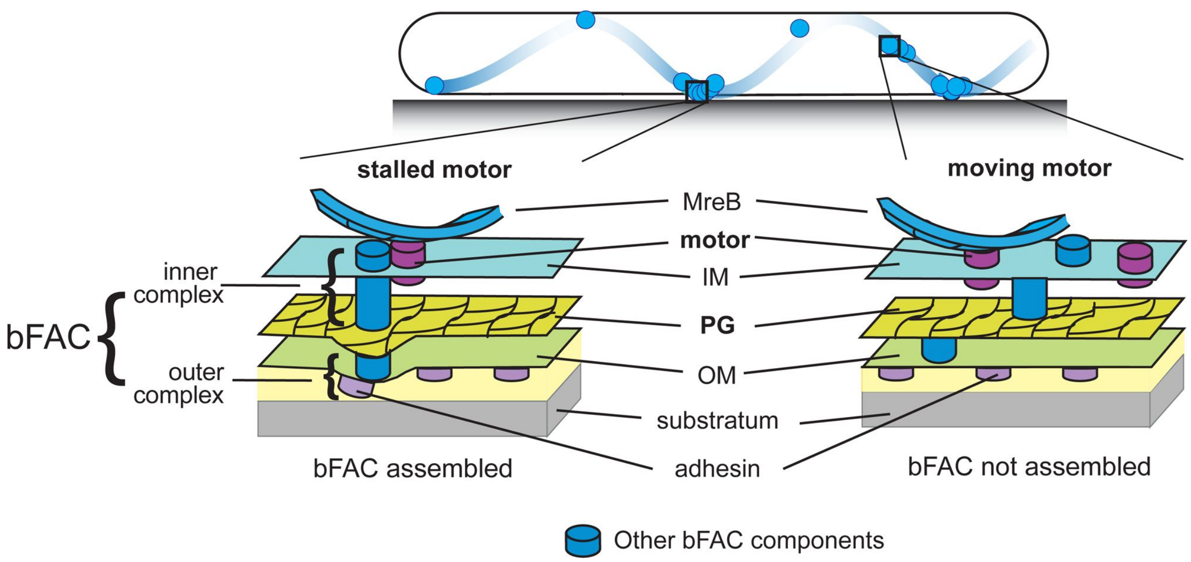

Fig. 1.

Stationary bFACs drive *M. xanthus* gliding.

Motors carrying incomplete gliding complexes either diffuse or move rapidly along helical paths but do not generate propulsion. Motors stall and become nearly static relative to the substrate when they assemble into complete bFACs with other motor-associated proteins at the ventral side of the cell. Stalled motors push MreB and bFACs in opposite directions and thus exert force against outer membrane adhesins. Overall, as motors transport bFACs toward lagging cell poles, cells move forward but bFACs remain static relative to the substrate. IM, inner membrane; OM, outer membrane.

thus restores gliding motility. Hence, the LTG activity of AgmT anchors bFAC to PG, potentially by modifying PG structure. Our findings reveal the long-sought connection between PG and bFAC that allows mechanical force to transmit across the PG cell wall.

Results

AgmT, a putative LTG, is essential for gliding motility

To elucidate how bFACs interact with PG, we searched for potential PG-binding domains among the proteins that are required for gliding motility. A previous report identified 35 gliding-related genes in *M. xanthus* through transposon-mediated random mutagenesis[23]. Among these genes, *agmT* (ORF K1515_04910[24]) was predicted to encode a protein with a large "periplasmic solute-binding" domain[23]. After careful analysis, we found that AgmT showed significant similarity to the widely conserved YceG/MltG family LTGs. About 70% of bacterial genomes, including firmicutes, proteobacteria, and actinobacteria, encode YceG/MltG domains[25]. Especially, the putative active site, Glu223 (corresponding to E218 in *E. coli* MltG)[25], is conserved in AgmT (**Fig. 2A**). To confirm the function of AgmT in gliding, we constructed an *agmT* in-frame deletion mutant. To eliminate S-motility, we further knocked out the *pilA* gene that encodes pilin for type IV pilus. In contrast to the *pilA*$^{-}$ cells that were still motile with gliding motility, the Δ*aglR pilA*$^{-}$ cells that lack an essential component in the gliding motor, were unable to move outward from the colony edge. Similarly, the Δ*agmT pilA*$^{-}$ cells also formed sharp colony edges, indicating that they were nonmotile with gliding (**Fig. 2B**). As AgmT is a putative LTG for PG, we then tested if its predicted active site, Glu223, is required for gliding. Because the amino acid following Glu223 is also a glutamate (Glu224), we replaced both glutamate residues with alanine (AgmT^{EAEA}) using site-directed mutagenesis to alter their codons on the chromosome. *agmT*^{EAEA} *pilA*$^{-}$ cells were nonmotile (**Fig. 2B**). Thus, the putative LTG activity of AgmT is essential for *M. xanthus* gliding motility.

Besides *agmT*, the genome of *M. xanthus* contains 13 genes that encode putative LTGs (**Table S1**). To test if these proteins also contribute to gliding, we knocked out each of these 13 genes in the *pilA*$^{-}$ background. The resulting mutants all retained gliding motility, indicating that AgmT is the only LTG that is required for *M. xanthus* gliding (**Fig. S1**).

AgmT is an LTG

To determine if AgmT is an LTG, we expressed the periplasmic domain (amino acids 25 – 339) of wild-type AgmT and AgmT^{EAEA} in *E. coli*. We purified PG from wild-type *M. xanthus* cells, labeled it with Remazol brilliant blue (RBB), and tested if the purified AgmT variants hydrolyze labeled PG *in vitro* and release the dye[26,27]. Similar to lysozyme that specifically cleaves the ®-1,4-glycosidic bonds in PG, wild-type AgmT solubilized dye-labeled *M. xanthus* PG that absorbs light at 595 nm. In contrast, the AgmT^{EAEA} variant failed to release the dye (**Fig. 3A**). Hence, AgmT displays LTG activity *in vitro*.

Whereas AgmT does not affect growth rate (**Fig. S2**), cells that lacked AgmT or expressed AgmT^{EAEA} maintained rod shape but were slightly shorter and wider than the wild-type ones (**Fig. 3B**). Nevertheless, altered morphology alone does not likely account for abolished gliding. In fact, a previously reported mutant that lacks all three class A penicillin-binding proteins (*⊗3*) displays similarly shortened and widened morphology[28], but is still motile by gliding (**Fig. S1**).

A recent report revealed that *Vibrio cholerae* MltG degrades un-crosslinked PG turnover products and prevents their detrimental accumulation in the periplasm[29]. To test if AgmT plays a similar role in *M. xanthus*, we used mecillinam to induce cell envelope stress. Mecillinam is a ®-lactam that induces the production of toxic, un-crosslinked PG strands[30]. Rather than collapsing the rod shape, mecillinam only causes bulging near the centers of wild-type *M. xanthus* cells, whereas

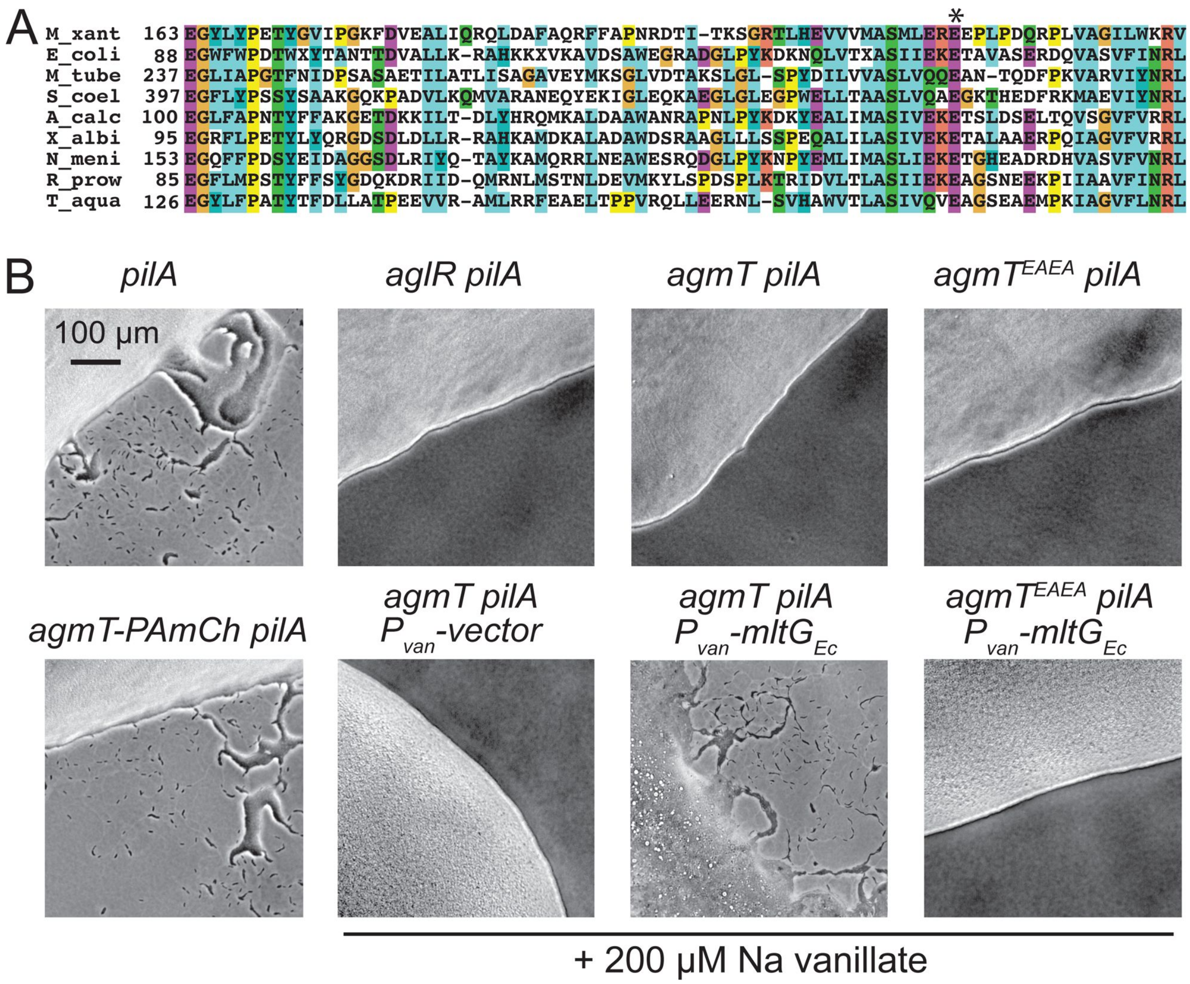

Fig. 2.

AgmT, a putative lytic transglycosylase, is required *for M. xanthus* gliding.

A) AgmT shows significant similarity to a widely conserved PG transglycosylase in YceG/MltG family. The conserved glutamine residue is marked by an asterisk. M_xant, *M. xanthus*; E_coli, *E. coli*; M_tube, *Mycobacterium tuberculosis*; S_coel, *Streptomyces coelicolor*; A_calc, *Acinetobacter calcoaceticus*; X_albi, *Xanthomonas albilineans*; N_meni, *Neisseria menningitidis*; R_prow, *Rickettsia prowazekii*; T_aqua, *Thermus aquaticus*. **B)** AgmT is required for *M. xanthus* gliding. Colony edges were imaged after incubating cells on 1.5% agar surface for 24 h. Deleting or Disabling the active site of AgmT abolishes gliding but fusing an PAmCherry (PAmCh) to its C-terminus does not. Heterologous Expression of *E. coli* MltG restores gliding of *⊗agmT* cells but not the cells that express $\mathrm{AgmT}^{\mathrm{EAEA}}$.

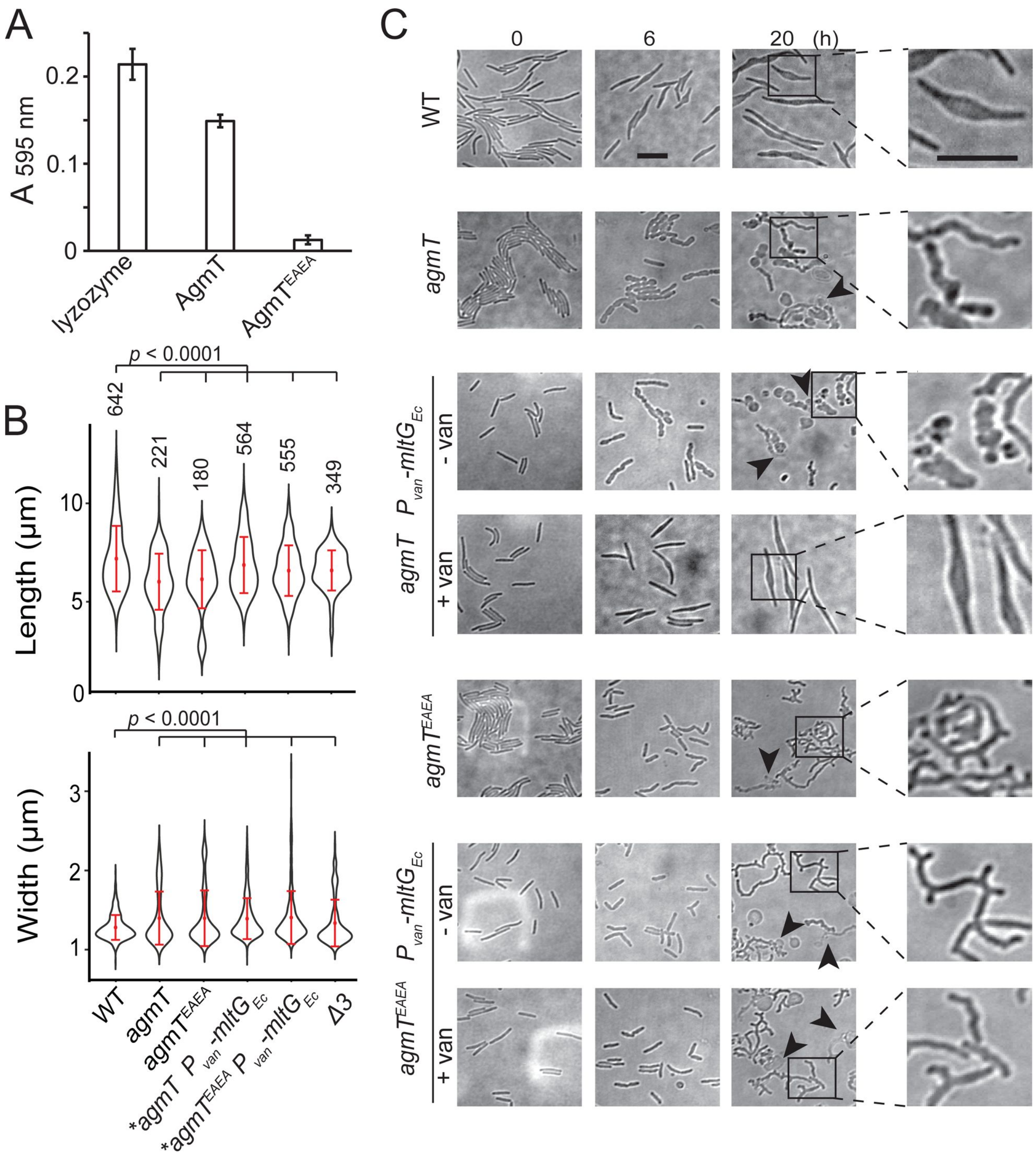

Fig. 3.

AgmT regulates cell morphology and integrity under antibiotic stress.

A) Purified AgmT solubilizes dye-labeled PG sacculi, but AgmT^{EAEA} does not. Lysozyme serves as a positive control. Absorption at 595 nm was measured after 18 h incubation at 25 °C. Data are presented as mean values ± SD from three replicates. **B)** AgmT regulates cell morphology. Compared to wild-type cells, cells that lack AgmT and express AgmT^{EAEA} are significantly shorter and wider. Heterologous expression of *E. coli* MltG partially restores cell length but not cell width in *agmT* and *agmT^{EAEA}* backgrounds. Asterisks, 200 ⌈M sodium vanillate added. *p* values were calculated using a one-way ANOVA test between two unweighted, independent samples. Whiskers indicate the 25th - 75th percentiles and red dots the median. The total number of cells analyzed is shown on top of each plot. **C)** AgmT regulates cell morphology and integrity under mecillinam stress (100 ⌈g/ml). Expressing *E. coli* MltG by a vanillate-inducible promoter (P_{van}) restores resistance against mecillinam in *agmT* cells but not in the cells that express AgmT^{EAEA}. van,. Arrows point to newly lysed cells. Scale bars, 5 µm.

large-scale cell lysis does not occur and cell poles still maintain rod shape even after prolonged (20 h) treatment (**Fig. 3C**)[28]. Thus, wild-type *M. xanthus* can largely mitigate mecillinam stress. In contrast, mecillinam-treated *agmT* cells increased their width drastically, displayed significant cell surface irregularity along their entire cell bodies, and lysed frequently (**Fig. 3C**). Cells expressing AgmT^{EAEA} as the sole source of AgmT displayed similar, but even stronger phenotypes under mecillinam stress, growing into elongated, twisted filaments that formed multiple cell poles and lysed frequently (**Fig. 3C**). These results confirm that similar to *V. cholerae* MltG, *M. xanthus* AgmT is important for maintaining cell integrity under mecillinam stress. The different responses from *agmT* and *agmT*^{EAEA} cells suggest that AgmT could interact with other PG-related enzymes and that the inactive AgmT variant might still modulate PG through these enzymes. Alternatively, other LTGs could partially substitute AgmT while AgmT^{EAEA} blocks these enzymes from accessing un-crosslinked PG strands.

AgmT is essential for bFACs assembly

How does AgmT support gliding? We first tested if AgmT regulates the function of the gliding motor. For this purpose, we quantified the motor dynamics by tracking single particles of AglR, an essential component of the motor. We spotted the cells that express a fully functional, photo-activatable mCherry (PAmCherry)-labeled AglR[13] on 1.5% agar surface. We used a 405-nm excitation laser to activate the fluorescence of a few labeled AglR particles randomly in each cell and quantified their localization using a 561-nm laser at 10 Hz using single particle tracking photo-activated localization microscopy (sptPALM) under highly inclined and laminated optical sheet (HILO) illumination[13,18,31]. Using this setting, only a thin section of each cell surface that was close to the coverslip was illuminated. To analyze the data, we only chose the fluorescent particles that remained in focus for 4 - 12 frames (0.4 - 1.2 s). As free PAmCherry particles diffuse extremely fast in the cytoplasm, entering and exiting the focal plane frequently, they usually appear as blurry objects that cannot be followed at 10-Hz close to the membrane[18]. For this reason, the noise from any potential degradation of AglR-PAmCherry was negligible. Consistent with our previous results[13,16], 32.1% (n = 2,700) of AglR-PAmCherry particles remained within one pixel (160 nm · 160 nm) before photobleach, indicating that they were nonmotile. The remaining 67.9% AglR-PAmCherry particles were motile, leaving trajectories of various lengths (**Fig. 4A**, **4B**).

bFAC assembly is sensitive to mechanical cues. As the agar concentration increases in gliding substrate, more motor molecules engage in bFACs, where they appear immobile[10,15]. We exposed *aglR-PAmCherry* cells to 405-nm excitation (0.2 kW/cm^2) for 2 s, where most PAmCherry molecules were photoactivated and used epifluorescence to display the overall localization of AglR[13,18,32]. As the agar concentration increased bFACs increased significantly in size (**Fig. 4C**). Accordingly, the immobile AglR particles detected by sptPALM increased from 10.7% (n = 1112) on 0.5% agar to 45.2% (n = 726) on 5.0% agar surface (**Fig. 4B**). Thus, the stationary population of AglR-PAmCherry particles reflects the AglR molecules in fully assembled bFACs.

In contrast to the cells that expressed wild-type AgmT (*agmT*$^+$), AglR particles were hyper motile in the cells that lacked AgmT or expressed AgmT^{EAEA}. On 1.5% agar surfaces, the immobile populations of AglR particles decreased to 10.5% (n = 2,989) and 11.7% (n = 1,408), respectively. Consistently, AglR-containing bFACs were not detectable in these cells, even on 5% agar surfaces (**Fig. 4A-C**).

To test if other components in the gliding machinery also depend on AgmT to assemble into bFACs, we tested the localization of AglZ, a cytoplasmic protein that is commonly used for assessing bFAC assembly[11,33]. In the cells that expressed wild-type AgmT, yellow fluorescent protein (YFP)-labeled AglZ formed a bright cluster at the leading cell pole and multiple stationary, near-evenly spaced clusters along the cell body, indicating the assembly of bFACs (**Fig. 4D**). In contrast, AglZ-YFP only formed single clusters at cell poles in the cells that lacked AgmT or expressed AgmT^{EAEA} (**Fig. 4D**). Taken together, the LTG activity of AgmT is essential for bFACs assembly.

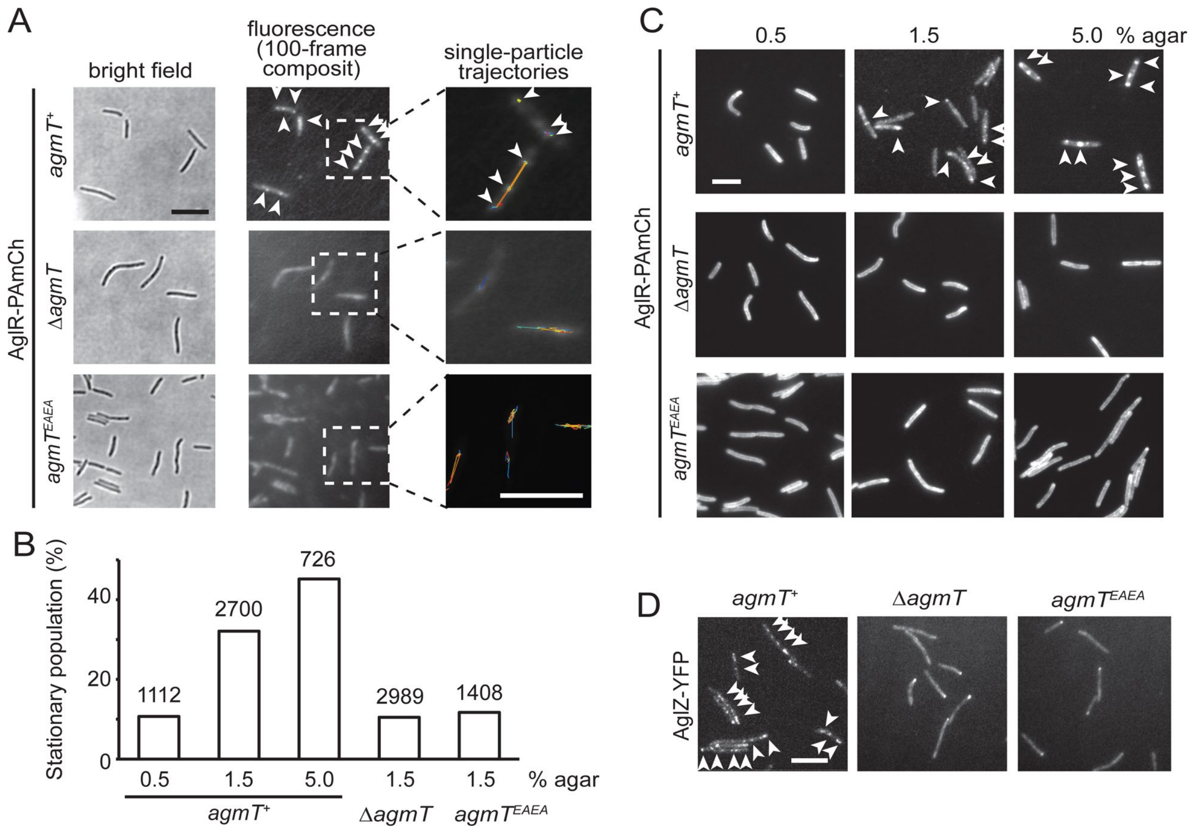

Fig. 4.

AgmT and its LTG activity are essential for bFAC assembly.

A) Overall distribution of AglR-PAmCherry particles on 1.5% agar surface is displayed using the composite of 100 consecutive frames. Single-particle trajectories of AglR-PAmCherry were generated from the same frames. Individual trajectories are distinguished by colors. Deleting *agmT* or disabling the transglycosylase activity of AgmT (AgmT^{EAEA}) decrease the stationary population of AglR particles. **B)** The stationary population of AglR-PAmCherry particles, which reflects the AglR molecules in bFACs, changes in response to substrate hardness (controlled by agar concentration) and the presence and function of AgmT. The total number of particles analyzed is shown on top of each plot. **C)** AglR fails to assemble into bFACs in the absence of active AgmT, even on 5.0% agar surfaces. **D)** AgmT and its LTG activity also support the assembly of AglZ into bFACs. White arrows point to bFACs. Scale bars, 5 μm.

AgmT does not assemble into bFACs

We first hypothesized that AgmT itself could assemble into bFACs, where it could either generate pores in PG that allow the inner and outer gliding complexes interact directly or bind to PG and recruit other components to bFACs through protein-protein interactions. If this is the case, AgmT should localize in bFACs. To test this possibility, we labeled AgmT with PAmCherry on its C-terminus and expressed the fusion protein as the sole source of AgmT using the native promoter and locus of *agmT*. *agmT-PAmCherry pilA*$^{-}$ cells displayed gliding motility that was indistinguishable from *pilA*$^{-}$ cells (**Fig. 2B**), indicating that the fusion protein is functional. Immunoblot using an anti-mCherry antibody showed that AgmT-PAmCherry accumulated in two bands that corresponded to monomers and dimers of the full-length fusion protein, respectively (**Fig. 5A**). This result is consistent with the structure of *E. coli* MltG that functions as homodimers (PDB: 2r1f). Importantly, AgmT-PAmCherry was about 50 times more abundant than AglR-PAmCherry (**Fig. 5A**). To test if AgmT itself assembles into bFACs, we first expressed AgmT-PAmCherry and AglZ-YFP together using their respective loci and promoter. We exposed *agmT-PAmCherry* cells to 405-nm excitation (0.2 kW/cm^2) for 2 s to visualize the overall localization of AgmT. We found that on a 1.5% agar surface that favors gliding motility, AglZ aggregated into bFACs whereas AgmT localized near evenly along cell bodies without forming protein clusters (**Fig. 5B**). Thus, AgmT does not localize into bFACs. These results echo the fact that despite its abundance, AgmT has not been identified as a component in bFACs despite extensive pull-down experiments using various bFAC components as baits[6,10,34].

We reasoned that if AgmT assembles into bFACs, AgmT and gliding motors should display similar dynamic patterns. To test this possibility further, we used sptPALM to track the movements of AgmT-PAmCherry single particles on 1.5 agar surfaces at 10 Hz. Distinct from the AglR particles of which 32.1% remained stationary, AgmT moved in a diffusive manner, showing no significant immobile population. Importantly, compared to the diffusion coefficients (*D*) of mobile AglR particles ($1.8 \cdot 10^{-2} \pm 3.6 \cdot 10^{-3}$ ⌈m^2/s (n = 1,833)), AgmT particles diffused much faster ($D = 2.9 \cdot 10^{-2} \pm 5.3 \cdot 10^{-3}$ ⌈m^2/s (n = 8,548)). Taken together, neither the localization nor dynamics of AgmT displayed significant correlation with bFAC components. We thus conclude that AgmT does not assemble into bFACs

AgmT connects bFACs to PG through its LTG activity

We then explored the possibility that AgmT could modify PG through its LTG activity and thus generate the anchor sites for certain components in bFACs. If this hypothesis is true, heterologous expression of a non-native LTG could rescue gliding motility in the ⊗*agmT* strain. To test this hypothesis, we fused the *E. coli mltG* gene (*mltG*$_{Ec}$) to a vanillate-inducible promoter and inserted the resulting construct to the *cuoA* locus on *M. xanthus* chromosome that does not interfere gliding[35]. We subjected the ⊗*agmT* and *agmT*^{EAEA} cells that expressed MltG_{Ec} to mecillinam (100 ⌈g/ml) stress. Induced by 100 ⌈M sodium vanillate, MltG_{Ec} restored cell morphology and integrity of the ⊗*agmT* strain to the wild-type level (**Fig. 3C**). This result confirmed that MltG_{Ec} was expressed and enzymatically active in *M. xanthus*. In contrast, MltG_{Ec} failed to confer mecillinam resistance to the *agmT*^{EAEA} cells (**Fig. 3C**). A potential explanation is that AgmT^{EAEA} could still bind to PG and thus block MltG_{Ec} from accessing *M. xanthus* PG. Consistent with its activity, MltG_{Ec} restored gliding motility in the *agmT pilA* cells but not in the *agmT*^{EAEA} *pilA* ones (**Fig. 2B**). As MltG_{Ec} substitutes AgmT in gliding motility, it is the LTG activity, rather than specific interactions between AgmT and bFACs, that is required for gliding.

It is challenging to visualize how AgmT facilitates bFAC assembly through its enzymatic activity *in vitro*. First, the inner complex in a bFAC spans the cytoplasm, inner membrane, and periplasm and contains multiple proteins whose activities are interdependent[8,10,11]. It is hence difficult to purify the entire inner complex in its functional state. Second, Faure *et al.* hypothesized that AglQ and AglS in the gliding motor could bind PG directly[11]. However, such binding has not been

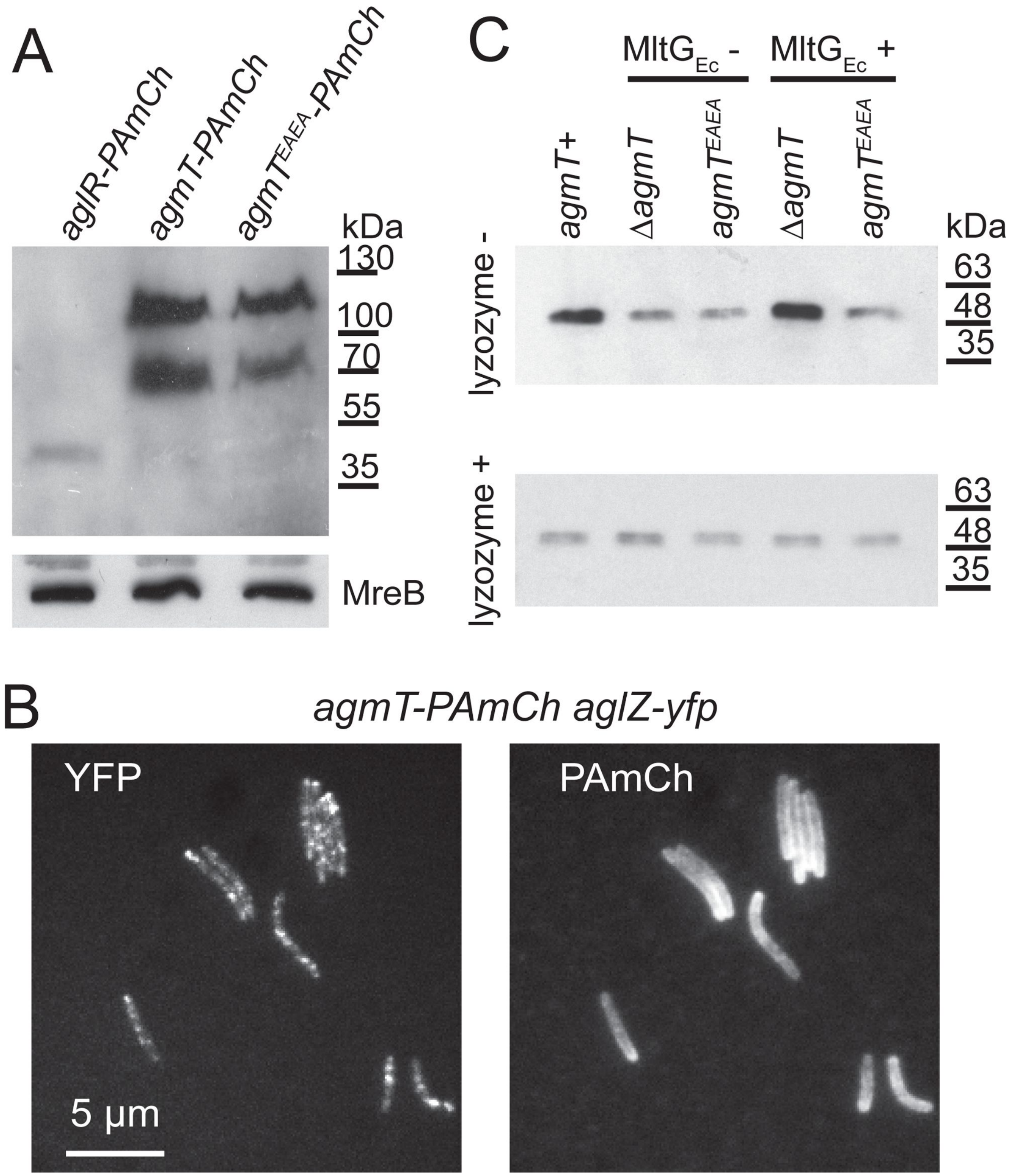

Fig. 5.

AgmT does not assemble into bFACs but connects bFACs to PG.

A) Immunoblotting using *M. xanthus* cell lysates and an anti-mCherry antibody shows that PAmCherry-labeled AgmT and AgmT^{EAEA} accumulate as full-length proteins. AgmT is significantly more abundant than the motor protein AglR. The bacterial actin homolog MreB visualized using an MreB antibody is shown as a loading control. **B)** AgmT does not aggregate into bFACs. **C)** The LTG activity of AgmT is required for connecting bFACs to PG and expressing the *E. coli* LTG MltG (MltG$_{Ec}$) restores PG binding by bFACs in cells that lack AgmT but not in the ones that express an inactive AgmT variant (AgmT^{EAEA}). AglR-PAmCherry was detected using an anti-mCherry antibody to mark the presence of bFACs that co-precipitate with PG-containing (lysozyme-) pellets. Lysates from the cells that express AglR-PAmCherry in different genetic backgrounds were pelleted by centrifugation in the presence and absence of lysozyme.

proved experimentally due to the difficulty in purifying the membrane integral AglRQS complex. To overcome these difficulties, we used AglR-PAmCherry to represent the inner complex of bFAC and investigated how bFACs bind PG in their native environment. To do so, we expressed AglR-PAmCherry in different genetic backgrounds as the sole source of AglR using the native *aglR* locus and promoter. We lysed these cells using sonication, subjected their lysates to centrifugation, isolated the pellets that contained PG, and detected the presence of AglR-PAmCherry in these pellets by immunoblots using a mCherry antibody. Eliminating AgmT and disabling its active site significantly reduced the amounts of AglR in the pellets (**Fig. 5B**). Because such pellets contain both the PG and membrane fractions, we further eliminated PG in cell lysates using lysozyme before centrifugation and determined the amount of AglR-PAmCherry in the pellets that only contained membrane fractions. Regardless of the presence and activity of AgmT, comparable amounts of AglR precipitated in centrifugation pellets after lysozyme treatment, indicating that AgmT does not affect the expression or stability of AglR (**Fig. 5B**). Thus, active AgmT facilitates the association between bFACs and PG. Strikingly, the heterologous expression of $\mathrm{MltG_{Ec}}$ enriched AglR-PAmCherry in the PG-containing pellets from the ⊗*agmT* cells but not the *agmT*$^{\mathrm{EAEA}}$ ones (**Fig. 5B**). These results indicate that AgmT connects bFACs to PG through its LTG activity.

Discussion

Gliding bacteria adopt a broad spectrum of nanomachineries. Compared to the trans-envelope secretion system that drives gliding in *Flavobacterium johnsoniae* and *Capnocytophaga gingivalis* that both contain PG, the gliding machinery of *M. xanthus* appears to lack a stable structure that transverses cell envelope, especially across the PG layer[2,36,37]. In order to generate mechanical force from the fluid *M. xanthus* gliding machinery, the inner complex must establish solid contact with PG and push the outer complex to slide[11,21]. In this work, we discovered AgmT as the long-sought factor that connects bFACs to PG. Such connection not only satisfies the requirement for force generation but is also essential for bFAC assembly.

It is rather unexpected that AgmT itself does not assemble into bFACs. Because $\mathrm{MltG_{Ec}}$ substitutes AgmT in gliding, rather than interacting with bFAC components specifically, AgmT facilitates bFAC assembly through its LTG activity. A few possibilities could explain the function of AgmT. First, LTGs usually break glycan strands and produce unique anhydro caps on their ends[38–41]. Certain components in bFACs could specifically bind to such modified glycan strands in PG and further recruit other components. Second, AgmT could generate small openings all over the PG meshwork that allow the inner and outer subcomplexes in bFACs to interact locally. Third, AgmT could reduce the overall rigidity of *M. xanthus* PG, allow the bulky inner subcomplexes to deform the PG layer, and thus establish a viscous coupling between bFACs and PG[6,21].

Then how does AgmT, a protein localized diffusively, facilitate bFAC assembly into near-evenly spaced foci? The actin homolog MreB is the only protein in bFACs that displays quasiperiodic localization independent of other components[18,22]. *M. xanthus* MreB assembles into bFACs and positions the latter near evenly along the cell body[6,13,18,22,42]. MreB filaments change their orientation in accordance with local membrane curvatures and could hence respond to mechanical cues[43–46]. Strikingly, *M. xanthus* does assemble bFACs in response to substrate hardness (**Fig. 4C**) and assembled bFACs could distort the cell envelope, generating undulations on the cell surface[6,10,13,47,48]. Taken together, whereas AgmT potentially modifies the entire PG layer, it is still the quasiperiodic MreB filaments that determine the loci for bFAC assembly in response to the mechanical cues from the gliding substrate.

Most bacteria encode multiple LTGs that function in PG growth, remodeling, and recycling. Trans-envelope structures often require highly specialized LTGs to penetrate PG[49]. This work provides an example that macro bacterial machineries domesticate non-specialized LTGs for specialized functions. Other examples include MltD in the *Helicobacter pylori* flagellum and MltE in the *E. coli* type VI secretion system[50,51]. Both *M. xanthus* AgmT and *E. coli* MltG belong to the YceG family, which is the first identified LTG family that is conserved in both Gram-negative and positive bacteria[39]. LTGs in this family localize to the inner membrane and feature periplasmic transglycosylase domains[39]. Due to their unique localization, the YceG family LTGs are important for determining the length of nascent glycan strands and for degrading the toxic un-crosslinked strands from futile PG synthesis[29,52]. Due to the wide distribution of YceG family LTGs, it is challenging to study how the unique *M. xanthus* gliding machinery adopts AgmT in evolution. Nevertheless, the fact that AgmT is the only *M. xanthus* LTG that belongs to this family (**Table S2**) could partially explain why it is the only LTG that has been adopted by the gliding machinery. It will be interesting to investigate if other LTGs, once anchored to the inner membrane, could also facilitate bFAC assembly.

Methods

Strains and growth conditions

M. xanthus strains used in this study are listed in **Table S2**. Vegetative *M. xanthus* cells were grown in liquid CYE medium (10 mM MOPS pH 7.6, 1% (w/v) Bacto™ casitone (BD Biosciences), 0.5% yeast extract and 8 mM $MgSO_4$) at 32 °C, in 125-ml flasks with vigorous shaking, or on CYE plates that contains 1.5% agar. All genetic modifications on *M. xanthus* were made on the chromosome. Deletion and insertion mutants were constructed by electroporating *M. xanthus* cells with 4 µg of plasmid DNA. Transformed cells were plated on CYE plates supplemented with 100 µg/ml sodium kanamycin sulfate and 10 µg/ml tetracycline hydrochloride when needed. AgmT and $AgmT^{EAEA}$ were labeled with PAmCherry at their C-termini by fusing their gene to a DNA sequence that encodes PAmCherry through a KESGSVSSEQLAQFRSLD (AAGGAGTCCGGCTCCGTGTCCTCCGAGCAGCTGGCCCAGTTCCGCTCCCTGGA C) linker. All constructs were confirmed by PCR and DNA sequencing.

Immunoblotting

The expression and stability of PAmCherry-labeled proteins were determined by immunoblotting using an anti-mCherry antibody (Rockland Immunochemicals, Inc., Lot 46705) and a goat anti-Rabbit IgG (H+L) secondary antibody, HRP (Thermo Fisher Scientific, catalog # 31460). MreB was detected using an anti-MreB serum[22] and the same secondary antibody. The blots were developed with Pierce™ ECL Western Blotting Substrate (Thermo Fisher Scientific REF 32109) and MINI-MED 90 processor (AFP Manufacturing).

Gliding assay

Cells at 4 $\times 10^9$ colony formation units (cfu)/ml were spotted on CYE plates (5 µl/spot) containing 1.5% agar, incubated at 32 °C for 48 h. Colony edges were photographed using a Nikon Eclipse™ e600 microscope with a 10× 0.30 NA objective and an OMAX™ A3590U camera.

Protein expression and purification

DNA sequences encoding amino acids 25 – 339 of AgmT and $AgmT^{EAEA}$ were amplified by polymerase chain reaction (PCR) and inserted into the pET28a vector (Novogen) between the restriction sites of *Eco*RI and *Hind*III. The resulting plasmids were transformed into *E. coli* strain BL21(DE3). Transformed cells were cultured in 20 ml LB (Luria-Bertani) broth at 37 °C overnight and used to inoculate 1 L LB medium supplemented with 1.0% glucose. Protein expression was induced by 0.1 mM IPTG (isopropyl-h-d-thiogalactopyranoside) when the culture reached an OD_{600} of 0.8. Cultivation was continued at 16 °C for 10 h before the cells were harvested by centrifugation at 6,000 × g for 20 min. Proteins were purified using the NGC™ Chromatography System (BIO-RAD) and a 5-ml HisTrap™ (Cytiva)[17,53]. Purified proteins were concentrated using Amicon™ Ultra centrifugal filter units (Millipore Sigma) with a molecular cutoff of 10 kDa and stored at −80 °C.

RBB assay

PG was purified following the published protocol[28,54]. In brief, *M. xanthus* cells were grown until mid-stationary phase and harvested by centrifugation (6,000 × g, 20 min, 25 °C). Supernatant was discarded and the pellet was resuspended and boiled in 1× PBS with 5% SDS for 2 h. SDS was removed by repetitive wash with MilliQ water and centrifugation (21,000 × g, 10 min, 25 °C). Purified PG from 100 ml culture was suspended into 1 ml 1× PBS and stored at −20 °C. RBB labelling of PG was performed essentially as previously described[26,27]. Purified sacculi were incubated with 20 mM RBB in 0.25 M NaOH overnight at 37 °C. Reactions were neutralized by adding equal volumes of 0.25 M HCl and RBB-labeled PG was collected by centrifugation at 21,000 × g for 15 min. Pellets were washed repeatedly with MilliQ water until the supernatants became colorless. RBB-labelled sacculi were incubated with purified AgmT and AgmT^{EAEA} (1 mg/ml) at 25 °C for 12 h. lysozyme (1 mg/ml) was used as a positive control. Dye release was quantified by the absorption at 595 nm from the supernatants after centrifugation (21,000 × g, 10 min, 25 °C).

PG binding assay

M. xanthus cells expressing AglR-PAmCherry were grown in liquid CYE to OD_{600} ~1, harvested by centrifugation (6,000 × g, 20 min, 25 °C), washed by 1× PBS, and resuspended into 1× PBS to OD_{600} 6. Cells (1 ml) were lysed using a Cole-Parmer 4710 Ultrasonic Homogenizer (40% output). Unbroken cells and large debris were eliminated by centrifugation (6,000 × g, 10 min, 25 °C). Supernatants were subjected to centrifugation at 21,000 × g for 15 min. Pellets that contain the PG and membrane fractions were resuspended to 1 ml in 1× PBS. To collect the pellets that do not contain PG, supernatants from sonication lysates were incubated with 5 mg/ml lysozyme at 25 °C for 5 h before centrifugation. Resuspended pellets (5 ⌠l each) were mixed with equal volumes of 2× loading buffer and applied to SDS PAGE and immunoblotting.

Microscopy Analysis

For all imaging experiments, 5 ⌠l cells were spotted on an agarose pad. For the treatments with inhibitors, inhibitors were added into both the cell suspension and agarose pads. The length and width of cells were determined from differential interference contrast (DIC) images using a MATLAB (MathWorks) script[18,28,55]. DIC and fluorescence images of cells were captured using a Hamamatsu ImagEM X2™ EM-CCD camera C9100-23B (effective pixel size 160 nm) on an inverted Nikon Eclipse-Ti™ microscope with a 100× 1.49 NA TIRF objective. For sptPALM, *M. xanthus* cells were grown in CYE to 4 ×10^8 cfu/ml and PAmCherry was activated using a 405-nm laser (0.3 kW/cm^2), excited and imaged using a 561-nm laser (0.2 kW/cm^2). Images were acquired at 10 Hz. For each sptPALM experiment, single PAmCherry particles were localized in at least 100 individual cells from three biological replicates. sptPALM data were analyzed using a MATLAB (MathWorks) script[18,28,55]. Briefly, cells were identified using differential interference contrast images. Single PAmCherry particles inside cells were fit by a symmetric 2D Gaussian function, whose center was assumed to be the particle's position[18]. Particles that explored areas smaller than 160 nm × 160 nm (within one pixel) in 0.4 - 1.2 s were considered immobile. Sample trajectories were generated using the TrackMate[56] plugin in the ImageJ suite (*https://imagej.net*).

Acknowledgements

This work is supported by the National Institute of Health R01GM129000 to B. N.. C. R. C. is supported in the 2023 – 2024 academic year by a National Institute of Health Diverse Predoctoral Training in Genetics grant T32GM135115 awarded to the Genetics and Genomics Interdisciplinary Program at Texas A&M University.

Author Contributions

C. R. C., O. G. F., and B. N. designed the study, performed experiments and data analysis, and wrote the manuscript. All authors read and approved the manuscript.

Competing Interests

The authors declare no competing interests.

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Reviewer #1 (Public Review):

Summary:

This manuscript nicely outlines a conceptual problem with the bFAC model in A-motility, namely, how is the energy produced by the inner membrane AglRQS motor transduced through the cell wall into mechanical force on the cell surface to drive motility? To address this, the authors make a significant contribution by identifying and characterizing a lytic transglycosylase (LTG) called AgmT. This work thus provides clues and a future framework work for addressing mechanical force transmission between the cytoplasm and the cell surface.

Strengths:

(1) Convincing evidence shows AgmT functions as an LTG and, surprisingly, that mltG from E. coli complements the swarming defect of an agmT mutant.

(2) Authors show agmT mutants develop morphological changes in response to treatment with a β-lactam antibiotic, mecillinam.

(3) The use of single-molecule tracking to monitor the assembly and dynamics of bFACs in WT and mutant backgrounds.

(4) The authors understand the limitations of their work and do not overinterpret their data.

Weaknesses:

(1) A clear model of AgmT's role in gliding motility or interactions with other A-motility proteins is not provided. Instead, speculative roles for how AgmT enzymatic activity could facilitate bFAC function in A-motility are discussed.

(2) Although agmT mutants do not swarm, in-depth phenotypic analysis is lacking. In particular, do individual agmT mutant cells move, as found with other swarming defective mutants, or are agmT mutants completely nonmotile, as are motor mutants?

(3) The bioinformatic and comparative genomics analysis of agmT is incomplete. For example, the sequence relationships between AgmT, MltG, and the 13 other LTG proteins in M. xanthus are not clear. Is E. coli MltG the closest homology to AgmT? Their relationships could be addressed with a phylogenetic tree and/or sequence alignments. Furthermore, are there other A-motility genes in proximity to agmT? Similarly, does agmT show specific co-occurrences with the other A-motility genes across genera/species?

(4) Related to iii, what about the functional relationship of the endogenous 13 LTG genes? Although knockout mutants were shown to be motile, presumably because AgmT is present, can overexpression of them, similar to E. coli MltG, complement an agmT mutant? In other

words, why does MltG complement and the endogenous LTG proteins appear not to be relevant?

(5) Based on Figure 2B, overexpression of MltG enhances A-motility compared to the parent strain and the agmT-PAmCh complemented strain, is this actually true? Showing expanded swarming colony phenotypes would help address this question.

(6) Cell flexibility is correlated with gliding motility function in M. xanthus. Since AgmT has LTG activity, are agmT mutants less flexible than WT cells and is this the cause of their motility defect?

https://doi.org/10.7554/eLife.99273.1.sa1

Reviewer #2 (Public Review):

The manuscript by Carbo et al. reports a novel role for the MltG homolog AgmT in gliding motility in M. xanthus. The authors conclusively show that AgmT is a cell wall lytic enzyme (likely a lytic transglycosylase), its lytic activity is required for gliding motility, and that its activity is required for proper binding of a component of the motility apparatus to the cell wall. The data are generally well-controlled. The marked strength of the manuscript includes the detailed characterization of AgmT as a cell wall lytic enzyme, and the careful dissection of its role in motility. Using multiple lines of evidence, the authors conclusively show that AgmT does not directly associate with the motility complexes, but that instead its absence (or the overexpression of its active site mutant) results in the failure of focal adhesion complexes to properly interact with the cell wall.

An interpretive weakness is the rather direct role attributed to AgmT in focal adhesion assembly. While their data clearly show that AgmT is important, it is unclear whether this is the direct consequence of AgmT somehow promoting bFAC binding to PG or just an indirect consequence of changed cell wall architecture without AgmT. In E. coli, an MltG mutant has increased PG strain length, suggesting that M. xanthus's PG architecture may likewise be compromised in a way that precludes AglR binding to the cell wall. However, this distinction would be very difficult to establish experimentally. MltG has been shown to associate with active cell wall synthesis in E.c oli in the absence of protein-protein interactions, and one could envision a similar model in M. xanthus, where active cell wall synthesis is required for focal adhesion assembly, and MltG makes an important contribution to this process.

https://doi.org/10.7554/eLife.99273.1.sa0